

BALAJI DILIPSINGH RAJPUT

Biotechnology Professional | Bioinformatics Specialist | Research & QA/QC

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PROFESSIONAL SUMMARY

Results-driven **Biotechnology Professional** with hands-on experience in **Bioinformatics, Laboratory Research, and Quality Assurance (QA/QC)**. Skilled in **Python** (Biopython, Pandas, NumPy), **R** (Bioconductor), genomic data analysis, and wet-lab techniques including **PCR, gel electrophoresis, fermentation, and cell culture**. Proven track record of **optimizing laboratory workflows**, achieving **95% accuracy** in sample processing, and improving **bioethanol yield by 15%** through process optimization and statistical analysis. Seeking challenging roles in **Bioinformatics, Research, or QA/QC** in the pharmaceutical and biotechnology industry. **Immediate availability | Open to relocation across India.**

CORE TECHNICAL SKILLS

Biotechnology Laboratory

PCR, Gel Electrophoresis, Cell Culture, Protein Expression, Microbial Culture Management, Fermentation Methods, Recombinant DNA Technology, Spectrophotometry, High-Throughput Screening (HTS)

Bioinformatics & Programming

Python (Biopython, Pandas, NumPy, Matplotlib), R (Bioconductor), BLAST, ClustalW, Multiple Sequence Alignment, Phylogenetic Analysis, NGS Data Interpretation, NCBI, UniProt, Ensembl

Data Analysis & Statistics

SPSS, MATLAB, Regression Analysis, ANOVA, Hypothesis Testing, T-tests, Experimental Design, Data Visualization, Statistical Validation of Experimental Results

Quality Assurance & Compliance

QA/QC Process Optimization, Good Laboratory Practice (GLP) Compliance, SOP Documentation, Analytical Testing, Data Accuracy Protocols, Documentation & Reporting Standards

PROFESSIONAL EXPERIENCE

Bioinformatics Intern

2024 • 3 Months

Biotechnika | Remote

- Completed comprehensive **bioinformatics training** focusing on genomic data analysis and computational biology methodologies.
- Performed **DNA and protein sequence analysis** using BLAST, ClustalW, and phylogenetic tools for comparative genomics studies.
- Applied **Python (Biopython, Pandas, NumPy)** and **R (Bioconductor)** for biological data processing, visualization, and exploratory analysis.
- Worked extensively with **NCBI, UniProt, and Ensembl** for sequence retrieval, annotation, and functional interpretation.
- Interpreted **Next-Generation Sequencing (NGS)** data and contributed to development of automated workflows for sequence alignment.

Laboratory Intern

2023 • 2 Months

Gujarat Biotech Innovations | Vadodara, Gujarat

- Contributed to **biotechnology research operations and Quality Control (QC)** processes in a professional laboratory environment.

Executed core laboratory techniques: **PCR, gel electrophoresis, microbial culture management** with approximately 95% accuracy in sample processing.

RESEARCH PROJECTS

- Improved **laboratory efficiency by 20%** through workflow optimization and process streamlining initiatives. *Lead Researcher | 2024*
- Maintained detailed **GLP-compliant documentation** and enhanced data accuracy protocols for experimental records.
- Increased bioethanol yield by 15% through systematic microbial process optimization and fermentation parameter adjustment.
- Operated laboratory equipment including spectrophotometers, centrifuges, PCR reactors, and analysis software with precision and safety.
- Awarded **Best Research Project Award - Parul Institute (2024)** for outstanding research contribution.

Antimicrobial Nanoparticle Synthesis

Project Lead | 2024

- Synthesized silver nanoparticles using *Pseudomonas* spp. for pharmaceutical and antimicrobial applications.
- Demonstrated antimicrobial efficacy against multiple bacterial strains; explored potential applications in reducing antibiotic resistance.

Microbial Fuel Cell Development

Research Contributor | 2023

- Enhanced energy output by **30%** through innovative wastewater treatment methodologies and electrode optimization.
- Designed electrode configurations for optimized microbial activity and improved power generation efficiency.

Recombinant Protein Expression in *E. coli*

Research Assistant | 2023

- Achieved high-yield production of target proteins in *E. coli* expression systems for biotechnology and therapeutic research applications.

BIOINFORMATICS PORTFOLIO (GITHUB PROJECTS)

DNA Sequence Analyzer (Python)

Developed comprehensive tool for automated DNA sequence analysis from FASTA files with features including GC content calculation, nucleotide composition analysis, and sequence transformations (complement, reverse complement). Batch processing support with command-line interface.

Technologies: Python 3, Biopython, SeqIO module

Protein Structure Prediction Pipeline (R)

Built automated pipeline for protein secondary structure prediction and physicochemical property analysis including molecular weight calculation, isoelectric point, instability index, and hydrophobicity analysis.

Technologies: R Programming, Bioconductor, ProtParam toolkit

COVID-19 Variant Analysis Tool (Python)

Created comparative genomics tool for analyzing SARS-CoV-2 variants and mutation patterns with automated sequence retrieval from NCBI GenBank, variant comparison, and mutation hotspot identification.

Technologies: Python, Pandas, Matplotlib, Seaborn, NCBI E-utilities API

EDUCATION

Diploma in Biotechnology

Parul Institute of Technology, Vadodara | 2022 - 2025 | CGPA: 6.7/10

Academic Achievements: Dean's List for Academic Excellence (2024-2025), Top 10% Ranking in Molecular Biology Coursework (2024)

Relevant Coursework: Molecular Biology, Genetics, Microbiology, Biochemistry, Cell Biology, Immunology, Bioinformatics, Biostatistics, Fermentation Technology, Recombinant DNA Technology

CERTIFICATIONS

- Advanced Biotechnology Techniques - Parul University (2025)
- Bioinformatics Training Certification - Biotechnika (2024)
- Digital Marketing Certification - ITI Vadodara (2021)

ACHIEVEMENTS & RECOGNITION

- Best Research Project Award:** Bioethanol Production Optimization - Parul Institute (2024)
- Dean's List Recognition:** Academic Excellence (2024-2025)

- **Top 10% Ranking:** Molecular Biology Coursework - Parul Institute (2024)

KEY STRENGTHS

- ✓ Analytical & Critical Thinking
- ✓ Laboratory Safety Compliance
- ✓ Problem-Solving Mindset
- ✓ Cross-Functional Collaboration
- ✓ Process Optimization
- ✓ Quality-Focused Approach
- ✓ Research Documentation
- ✓ Data-Driven Decision Making
- ✓ Technical Report Writing
- ✓ Attention to Detail
- ✓ Quick Learner & Adaptable
- ✓ Time Management

ADDITIONAL INFORMATION

Date of Birth: 20 June 2001

Languages: English (Fluent), Hindi (Fluent), Gujarati (Native)

Availability: Immediate

Relocation: Open to anywhere in India

Research Interests: Pharmaceutical QA, Applied Data Analysis in Biology, Sustainable Biotechnology, Nanobiotechnology, Computational Biology

Volunteering: Active participation in local sustainability initiatives, Member of Parul University Biotechnology Club, Attended research symposiums and scientific presentations

Portfolio: github.com/balajirajput970 | linkedin.com/in/balajirajput-483a86194